

STAGE 1
trained, then frozen

V_t, L, S_t

wrench history $\mathcal{F}_t$

causal encoder z_t^F

Force-conditioned teacher T_θ

STAGE 2
only ϕ trained

Missing-force mode $T_{\theta, \phi}^{\text{miss}}$
token $z_{\emptyset F}$, adapter ϕ

STAGE 3
schedule replay (t, k)

Shared cached prefix and noise E_k, ϵ_k

force-conditioned (S_t, z_t^F)
 $\mathcal{P}(A_{T,t|k}^{\text{cond}})$

learned missing-force $(S_t, z_{\emptyset F})$
 $\mathcal{P}(A_{T,t|k}^{\text{miss}})$

query- k cached reference
 $\mathcal{P}(A_k^{\text{ref}}(t_j))$

force residual target
 $\Delta A_{T,t|k}^{\text{force}}$

staleness target
 $\Delta A_{T,k \rightarrow t}^{\text{stale}}$

The diagram illustrates the Residual Policy and Command Composition module. It takes four inputs: Z_k^{intent} , $u_t^F = E_T^F(\mathcal{F}_t)$, S_t , and $A_k^{\text{ref}}(t)$. These inputs are processed by a Shared attention block Φ . The block outputs two heads: a force head $\Delta \hat{A}_{t,1:K}^{\text{force}}$ and a staleness head $\Delta \hat{A}_{t,1:K}^{\text{stale}}$. The force head output is used to calculate the cached action from update k , $\mathcal{P}(A_k^{\text{ref}}(t_j))$. The staleness head output is used to calculate the current S_t and cached S_k , $\Gamma(S_t, S_k)$. The force head output is also used to calculate the Stage-3 targets, $\mathcal{L}_{\text{distill}}$ on both heads. The final output is the Commanded pose $\mathcal{P}(A_{t,j}^{\text{cmd}})$, which is the sum of the cached action and the current S_t and cached S_k .